Exercise 2

SQL Aggregate functions & operators

1. List all distinct department in Students table

```
SELECT DISTINCT department
FROM student;
```

department
IT
HR
Finance

2. Average age per department

```
SELECT department, AVG (age) AS avg_age
FROM Student
GROUP By department;
```

department	avg-age
IT	21,5
HR	21
Finance	22

3. Show department with More than one Students

```
SELECT department, (OUNT(*) AS student_count
FROM Student
GROUP By department
HAVING (OUNT(* ) >1
```

department	Student-count
JI	2

4. Get all Students whose age is between 21 and 23

```
SELECT student_id, name, age
FROM Students
WHERE age BETWEEN 21 AND 23;
```

student_id	name	age
2	Bob	22
3	Charlie	21
4	Diana	23
5	Eve	22

5. List all student in the IT or HR department who are older than 21

```
SELECT student_id, name, age, department
FROM Students
WHERE department IN department IN ('IT', 'HR')
AND age > 21;
```

Student_Id	name	age	department
2	Bob	22	IT
4	Diana	23	HR

6. Show total credits per department, only for departments with more than 5 total credits.

```
SELECT department, SUM(credits) AS total_credit
FROM Courses
GROUP BY department
HAVING SUM(credits) > 5;
```

department	total_credits
IT	4

7. List all Course that do not have 4 credits

```
SELECT Course_Id; course_name, department, credits.
FROM Courses
WHERE credits = 24;
```

Course_id	course_name	department	credits
101	SQL basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	2

8, Show the top3 course by credit in descending order

```
SELECT course_id, course_name, department, credit
FROM courses
ORDER BY credit DESC
LIMIT 3;
```

course_id	course_name	department	credit
102	Python	IT	4
103	Data Science	IT	4
101	SQL Basics	IT	3

9. Get Maximum, minimum, and average grade across all enrollment

```
SELECT MAX(grade) AS max_grade,
       MIN(grade) AS min_grade
       AVG(grade) AS avg_grade
FROM enrollment;
```

Max_grade	Min_grade	avg_grade
90	78	84.33

10. Count how many enrollment exist per course

```
SELECT course_id; COUNT(*) AS enrollment_count
from enrollments
GROUP BY course_id
```

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

11. Find total salary and total bonus per department

```
SELECT department,
       SUM(salary) AS total_salary
       SUM(bonus) AS total_bonus
FROM salaries
GROUP By department;
```

department	total_salary	total_bonus
IT	122000	10500
HR	109000	8500
finance	70000	6000

12. Show department where average salary is above 55000

```
SELECT department,
       AVG(salary) AS avg_salary
FROM salaries
GROUP By department
HAVING AVG(salary) > 55000;
```

department	avg_salary
IT	61000
Finance	70000

13. List employees whose salary plus bonus is greater than 60000

```
SELECT employee_id, name, salary, bonus; SUM(salary + bonus) AS total_compensation
FROM Salaries
WHERE (salary + bonus) > 60000;
```

employee_id	name	salary	bonus	total compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

14 Show total and average budget per department only include department with average budget above 70 000.

```
SELECT department,
       SUM(budget) AS total_budget
       AVG(budget) AS avg_budget
FROM projects
GROUP by department
HAVING AVG(budget) > 70 000;
```

department	total_budget	Avg_budget
IT	270000	135 000
finance	80 000	80 000

15. List all projects with budgets between 50 000 and 120 000, excluding the marketing department

```
SELECT project_id,
       project_name,
       department,
       budget
FROM projects
WHERE budget BETWEEN 50 000 AND 120 000
AND department <> 'Marketing';
```

Project_id	project_name	department	budget
1	AI App	IT	120 000
2	Payroll system	finance	80 000
5	H/R portal	HR	50 000